

ID3

$$Info(D) = - \sum_{i=1}^m p_i \log_2(p_i)$$

$$Info_A(D) = \sum_{j=1}^v \frac{|D_j|}{|D|} \times Info(D_j)$$

$$Gain(A) = Info(D) - Info_A(D)$$

Dealing with Continuous-Valued Attributes

Temperature	40	48	60	72	80	90
PlayTennis	No	No	Yes	Yes	Yes	No

Information gain of Temperature_{>54}

Values (Temp_{>54}) = < 54, > 54

$$S = [3+, 3-] \quad Entropy(S) = -\frac{3}{6} \log_2 \frac{3}{6} - \frac{3}{6} \log_2 \frac{3}{6} = 1.0$$

$$S_{<54} \leftarrow [0+, 2-] \quad Entropy(S_{<54}) = 0.0$$

$$S_{>54} \leftarrow [3+, 1-] \quad Entropy(S_{>54}) = -\frac{3}{4} \log_2 \frac{3}{4} - \frac{1}{4} \log_2 \frac{1}{4} = 0.8113$$

$$Gain(S, Temp_{>54}) = Entropy(S) - \sum_{v \in \{<54, >54\}} \frac{|S_v|}{|S|} Entropy(S_v)$$

$$Gain(S, Temp_{>54}) = Entropy(S) - \frac{2}{6} Entropy(S_{<54}) - \frac{4}{6} Entropy(S_{>54})$$

$$= 0.4591$$

Dealing with Continuous-Valued Attributes

Temperature	40	48	60	72	80	90
PlayTennis	No	No	Yes	Yes	Yes	No

Information gain of $Temperature_{>85}$

Values ($Temp_{>85}$) = $\langle 85, > 85$

$$S = [3+, 3-] \quad Entropy(S) = -\frac{3}{6} \log_2 \frac{3}{6} - \frac{3}{6} \log_2 \frac{3}{6} = 1.0$$

$$S_{<85} \leftarrow [3+, 2-] \quad Entropy(S_{<85}) = -\frac{3}{5} \log_2 \frac{3}{5} - \frac{2}{5} \log_2 \frac{2}{5} = 0.971$$

$$S_{>85} \leftarrow [0+, 1-] \quad Entropy(S_{>85}) = 0.0$$

$$Gain(S, Temp_{>85}) = Entropy(S) - \sum_{v \in \{\langle 85, > 85 \rangle\}} \frac{|S_v|}{|S|} Entropy(S_v)$$

$$Gain(S, Temp_{>85}) = Entropy(S) - \frac{5}{6} Entropy(S_{<85}) - \frac{1}{6} Entropy(S_{>85})$$

$$= 0.1908$$

Dealing with Continuous-Valued Attributes

Temperature	40	48	60	72	80	90
PlayTennis	No	No	Yes	Yes	Yes	No

$$\textit{Gain}(S, \textit{Temp}_{>54}) = 0.4591$$

$$\textit{Gain}(S, \textit{Temp}_{>85}) = 0.1908$$

Day	Outlook	Temp	Humidity	Wind	PlayTennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

Attribute: Outlook

Values (Outlook) = Sunny, Overcast, Rain

$$S = [9+, 5-] \quad Entropy(S) = -\frac{9}{14} \log_2 \frac{9}{14} - \frac{5}{14} \log_2 \frac{5}{14} = 0.94$$

$$S_{Sunny} \leftarrow [2+, 3-] \quad Entropy(S_{Sunny}) = -\frac{2}{5} \log_2 \frac{2}{5} - \frac{3}{5} \log_2 \frac{3}{5} = 0.971$$

$$S_{Overcast} \leftarrow [4+, 0-] \quad Entropy(S_{Overcast}) = -\frac{4}{4} \log_2 \frac{4}{4} - \frac{0}{4} \log_2 \frac{0}{4} = 0$$

$$S_{Rain} \leftarrow [3+, 2-] \quad Entropy(S_{Rain}) = -\frac{3}{5} \log_2 \frac{3}{5} - \frac{2}{5} \log_2 \frac{2}{5} = 0.971$$

$$Gain(S, Outlook) = Entropy(S) - \sum_{v \in \{Sunny, Overcast, Rain\}} \frac{|S_v|}{|S|} Entropy(S_v)$$

Gain(S, Outlook)

$$\begin{aligned} &= Entropy(S) - \frac{5}{14} Entropy(S_{Sunny}) - \frac{4}{14} Entropy(S_{Overcast}) \\ &\quad - \frac{5}{14} Entropy(S_{Rain}) \end{aligned}$$

$$Gain(S, Outlook) = 0.94 - \frac{5}{14} 0.971 - \frac{4}{14} 0 - \frac{5}{14} 0.971 = 0.2464$$

Attribute: Temp

Values (Temp) = Hot, Mild, Cool

$$S = [9+, 5-] \quad Entropy(S) = -\frac{9}{14} \log_2 \frac{9}{14} - \frac{5}{14} \log_2 \frac{5}{14} = 0.94$$

$$S_{Hot} \leftarrow [2+, 2-] \quad Entropy(S_{Hot}) = -\frac{2}{4} \log_2 \frac{2}{4} - \frac{2}{4} \log_2 \frac{2}{4} = 1.0$$

$$S_{Mild} \leftarrow [4+, 2-] \quad Entropy(S_{Mild}) = -\frac{4}{6} \log_2 \frac{4}{6} - \frac{2}{6} \log_2 \frac{2}{6} = 0.9183$$

$$S_{Cool} \leftarrow [3+, 1-] \quad Entropy(S_{Cool}) = -\frac{3}{4} \log_2 \frac{3}{4} - \frac{1}{4} \log_2 \frac{1}{4} = 0.8113$$

$$Gain(S, Temp) = Entropy(S) - \sum_{v \in \{Hot, Mild, Cool\}} \frac{|S_v|}{|S|} Entropy(S_v)$$

Gain(S, Temp)

$$\begin{aligned} &= Entropy(S) - \frac{4}{14} Entropy(S_{Hot}) - \frac{6}{14} Entropy(S_{Mild}) \\ &\quad - \frac{4}{14} Entropy(S_{Cool}) \end{aligned}$$

$$= 0.0289$$

Attribute: Humidity

Values (Humidity) = High, Normal

$$S = [9+, 5-] \quad Entropy(S) = -\frac{9}{14} \log_2 \frac{9}{14} - \frac{5}{14} \log_2 \frac{5}{14} = 0.94$$

$$S_{High} \leftarrow [3+, 4-] \quad Entropy(S_{High}) = -\frac{3}{7} \log_2 \frac{3}{7} - \frac{4}{7} \log_2 \frac{4}{7} = 0.9852$$

$$S_{Normal} \leftarrow [6+, 1-] \quad Entropy(S_{Normal}) = -\frac{6}{7} \log_2 \frac{6}{7} - \frac{1}{7} \log_2 \frac{1}{7} = 0.5916$$

$$Gain(S, Humidity) = Entropy(S) - \sum_{v \in \{High, Normal\}} \frac{|S_v|}{|S|} Entropy(S_v)$$

$$Gain(S, Humidity)$$

$$= Entropy(S) - \frac{7}{14} Entropy(S_{High}) - \frac{7}{14} Entropy(S_{Normal})$$

$$Gain(S, Humidity) = 0.94 - \frac{7}{14} 0.9852 - \frac{7}{14} 0.5916 = 0.1516$$

Attribute: Wind

Values (Wind) = Strong, Weak

$$S = [9+, 5-] \quad Entropy(S) = -\frac{9}{14} \log_2 \frac{9}{14} - \frac{5}{14} \log_2 \frac{5}{14} = 0.94$$

$$S_{Strong} \leftarrow [3+, 3-] \quad Entropy(S_{Strong}) = 1.0$$

$$S_{Weak} \leftarrow [6+, 2-] \quad Entropy(S_{Weak}) = -\frac{6}{8} \log_2 \frac{6}{8} - \frac{2}{8} \log_2 \frac{2}{8} = 0.8113$$

$$Gain(S, Wind) = Entropy(S) - \sum_{v \in \{Strong, Weak\}} \frac{|S_v|}{|S|} Entropy(S_v)$$

$$Gain(S, Wind) = Entropy(S) - \frac{6}{14} Entropy(S_{Strong}) - \frac{8}{14} Entropy(S_{Weak})$$

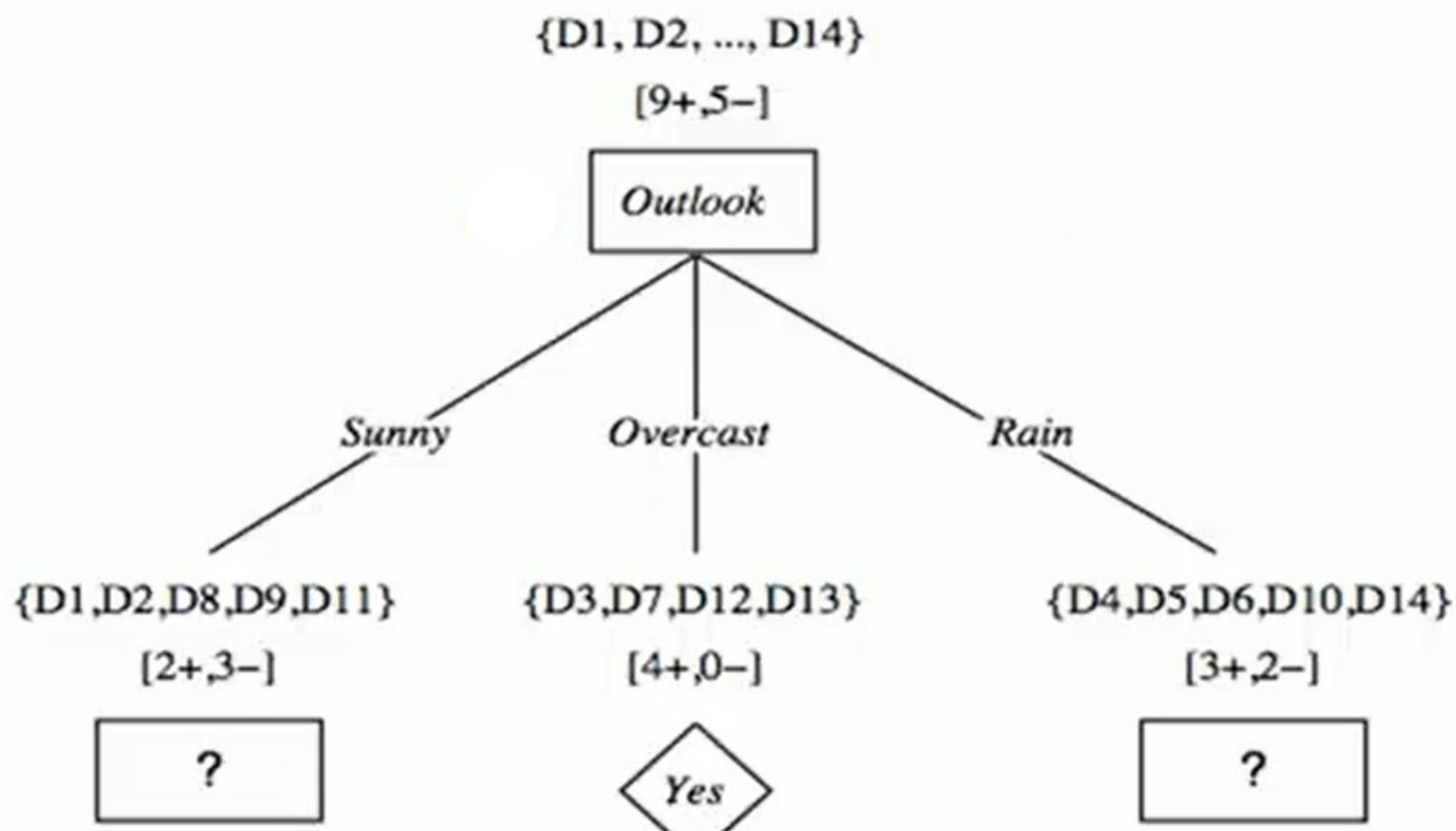
$$Gain(S, Wind) = 0.94 - \frac{6}{14} 1.0 - \frac{8}{14} 0.8113 = 0.0478$$

$$\textit{Gain}(S, \textit{Outlook}) = 0.2464$$

$$\textit{Gain}(S, \textit{Temp}) = 0.0289$$

$$\textit{Gain}(S, \textit{Humidity}) = 0.1516$$

$$\textit{Gain}(S, \textit{Wind}) = 0.0478$$



Day	Temp	Humidity	Wind	Play Tennis
D1	Hot	High	Weak	No
D2	Hot	High	Strong	No
D8	Mild	High	Weak	No
D9	Cool	Normal	Weak	Yes
D11	Mild	Normal	Strong	Yes

Attribute: Temp

Values (Temp) = Hot, Mild, Cool

$$S_{\text{Sunny}} = [2+, 3-] \quad \text{Entropy}(S_{\text{Sunny}}) = -\frac{2}{5} \log_2 \frac{2}{5} - \frac{3}{5} \log_2 \frac{3}{5} = 0.97$$

$$S_{\text{Hot}} \leftarrow [0+, 2-] \quad \text{Entropy}(S_{\text{Hot}}) = 0.0$$

$$S_{\text{Mild}} \leftarrow [1+, 1-] \quad \text{Entropy}(S_{\text{Mild}}) = 1.0$$

$$S_{\text{Cool}} \leftarrow [1+, 0-] \quad \text{Entropy}(S_{\text{Cool}}) = 0.0$$

$$\text{Gain}(S_{\text{Sunny}}, \text{Temp}) = \text{Entropy}(S) - \sum_{v \in \{\text{Hot}, \text{Mild}, \text{Cool}\}} \frac{|S_v|}{|S|} \text{Entropy}(S_v)$$

$$\text{Gain}(S_{\text{Sunny}}, \text{Temp})$$

$$= \text{Entropy}(S) - \frac{2}{5} \text{Entropy}(S_{\text{Hot}}) - \frac{2}{5} \text{Entropy}(S_{\text{Mild}})$$

$$- \frac{1}{5} \text{Entropy}(S_{\text{Cool}})$$

$$\text{Gain}(S_{\text{sunny}}, \text{Temp}) = 0.97 - \frac{2}{5} 0.0 - \frac{2}{5} 1 - \frac{1}{5} 0.0 = 0.570$$

Attribute: Humidity

Values (Humidity) = High, Normal

$$S_{Sunny} = [2+, 3-] \quad Entropy(S) = -\frac{2}{5} \log_2 \frac{2}{5} - \frac{3}{5} \log_2 \frac{3}{5} = 0.97$$

$$S_{High} \leftarrow [0+, 3-] \quad Entropy(S_{High}) = 0.0$$

$$S_{Normal} \leftarrow [2+, 0-] \quad Entropy(S_{Normal}) = 0.0$$

$$Gain(S_{Sunny}, Humidity) = Entropy(S) - \sum_{v \in \{High, Normal\}} \frac{|S_v|}{|S|} Entropy(S_v)$$

$$Gain(S_{Sunny}, Humidity) = Entropy(S) - \frac{3}{5} Entropy(S_{High}) - \frac{2}{5} Entropy(S_{Normal})$$

$$Gain(S_{Sunny}, Humidity) = 0.97 - \frac{3}{5} 0.0 - \frac{2}{5} 0.0 = 0.97$$

Attribute: Wind

Values (Wind) = Strong, Weak

$$S_{\text{Sunny}} = [2+, 3-] \quad \text{Entropy}(S) = -\frac{2}{5} \log_2 \frac{2}{5} - \frac{3}{5} \log_2 \frac{3}{5} = 0.97$$

$$S_{\text{Strong}} \leftarrow [1+, 1-] \quad \text{Entropy}(S_{\text{Strong}}) = 1.0$$

$$S_{\text{Weak}} \leftarrow [1+, 2-] \quad \text{Entropy}(S_{\text{Weak}}) = -\frac{1}{3} \log_2 \frac{1}{3} - \frac{2}{3} \log_2 \frac{2}{3} = 0.9183$$

$$\text{Gain}(S_{\text{Sunny}}, \text{Wind}) = \text{Entropy}(S) - \sum_{v \in \{\text{Strong}, \text{Weak}\}} \frac{|S_v|}{|S|} \text{Entropy}(S_v)$$

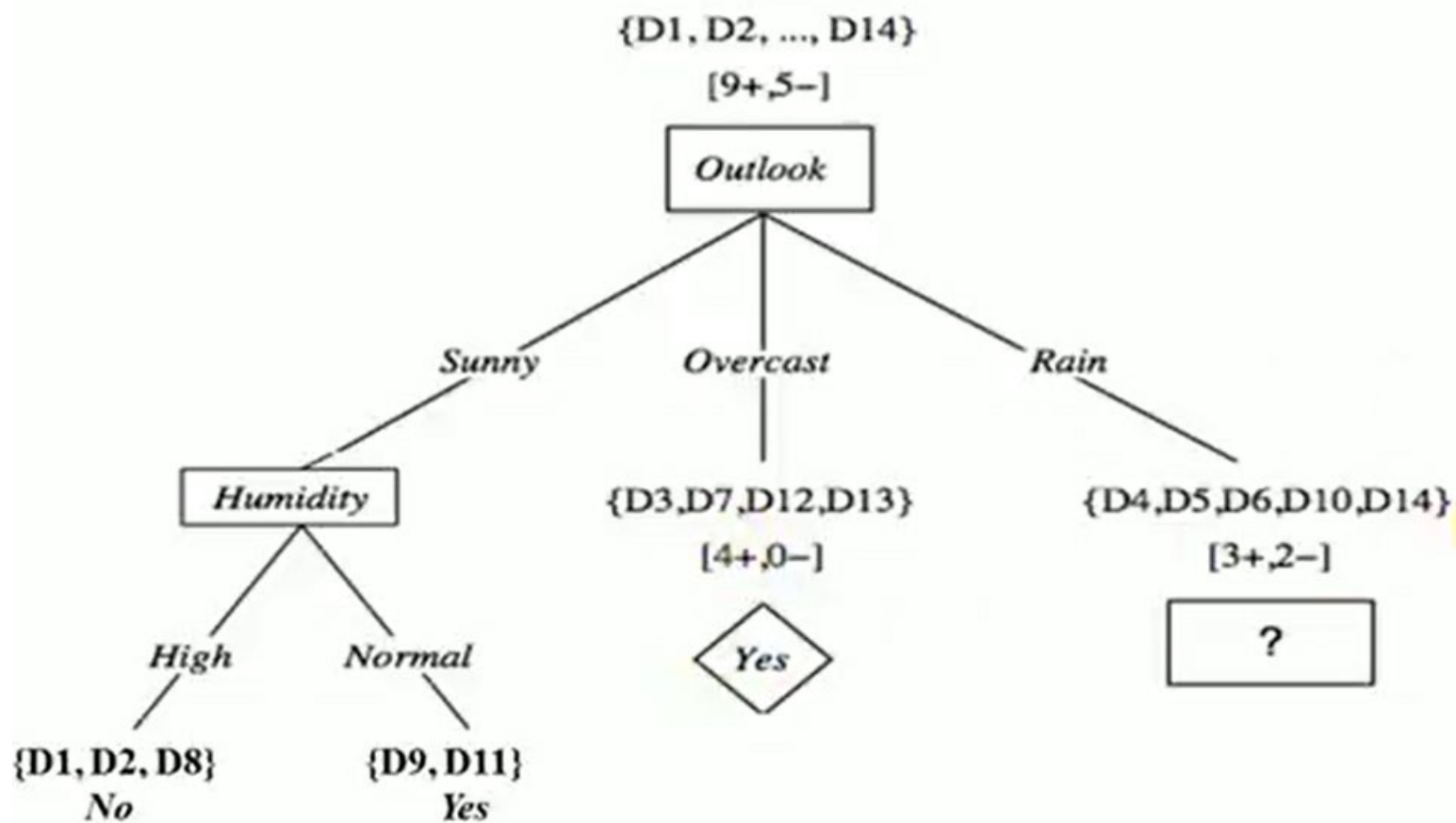
$$\text{Gain}(S_{\text{Sunny}}, \text{Wind}) = \text{Entropy}(S) - \frac{2}{5} \text{Entropy}(S_{\text{Strong}}) - \frac{3}{5} \text{Entropy}(S_{\text{Weak}})$$

$$\text{Gain}(S_{\text{Sunny}}, \text{Wind}) = 0.97 - \frac{2}{5} 1.0 - \frac{3}{5} 0.918 = 0.0192$$

$$\textit{Gain}(S_{\textit{sunny}}, \textit{Temp}) = 0.570$$

$$\textit{Gain}(S_{\textit{sunny}}, \textit{Humidity}) = 0.97$$

$$\textit{Gain}(S_{\textit{sunny}}, \textit{Wind}) = 0.0192$$



Day	Temp	Humidity	Wind	Play Tennis
D4	Mild	High	Weak	Yes
D5	Cool	Normal	Weak	Yes
D6	Cool	Normal	Strong	No
D10	Mild	Normal	Weak	Yes
D14	Mild	High	Strong	No

Attribute: Temp

Values (Temp) = Hot, Mild, Cool

$$S_{Rain} = [3+, 2-]$$

$$Entropy(S_{Sunny}) = -\frac{3}{5} \log_2 \frac{3}{5} - \frac{2}{5} \log_2 \frac{2}{5} = 0.97$$

$$S_{Hot} \leftarrow [0+, 0-]$$

$$Entropy(S_{Hot}) = 0.0$$

$$S_{Mild} \leftarrow [2+, 1-]$$

$$Entropy(S_{Mild}) = -\frac{2}{3} \log_2 \frac{2}{3} - \frac{1}{3} \log_2 \frac{1}{3} = 0.9183$$

$$S_{Cool} \leftarrow [1+, 1-]$$

$$Entropy(S_{Cool}) = 1.0$$

$$Gain(S_{Rain}, Temp) = Entropy(S) - \sum_{v \in \{Hot, Mild, Cool\}} \frac{|S_v|}{|S|} Entropy(S_v)$$

$$Gain(S_{Rain}, Temp)$$

$$= Entropy(S) - \frac{0}{5} Entropy(S_{Hot}) - \frac{3}{5} Entropy(S_{Mild})$$

$$- \frac{2}{5} Entropy(S_{Cool})$$

$$Gain(S_{Rain}, Temp) = 0.97 - \frac{0}{5} 0.0 - \frac{3}{5} 0.918 - \frac{2}{5} 1.0 = 0.0192$$

Attribute: Humidity

Values (Humidity) = High, Normal

$$S_{Rain} = [3+, 2-]$$

$$Entropy(S_{Sunny}) = -\frac{3}{5} \log_2 \frac{3}{5} - \frac{2}{5} \log_2 \frac{2}{5} = 0.97$$

$$S_{High} \leftarrow [1+, 1-]$$

$$Entropy(S_{High}) = 1.0$$

$$S_{Normal} \leftarrow [2+, 1-]$$

$$Entropy(S_{Normal}) = -\frac{2}{3} \log_2 \frac{2}{3} - \frac{1}{3} \log_2 \frac{1}{3} = 0.9183$$

$$Gain(S_{Rain}, Humidity) = Entropy(S) - \sum_{v \in \{High, Normal\}} \frac{|S_v|}{|S|} Entropy(S_v)$$

$$Gain(S_{Rain}, Humidity) = Entropy(S) - \frac{2}{5} Entropy(S_{High}) - \frac{3}{5} Entropy(S_{Normal})$$

$$Gain(S_{Rain}, Humidity) = 0.97 - \frac{2}{5} 1.0 - \frac{3}{5} 0.918 = 0.0192$$

Attribute: Wind

Values (wind) = Strong, Weak

$$S_{Rain} = [3+, 2-]$$

$$Entropy(S_{Sunny}) = -\frac{3}{5} \log_2 \frac{3}{5} - \frac{2}{5} \log_2 \frac{2}{5} = 0.97$$

$$S_{Strong} \leftarrow [0+, 2-]$$

$$Entropy(S_{Strong}) = 0.0$$

$$S_{Weak} \leftarrow [3+, 0-]$$

$$Entropy(S_{Weak}) = 0.0$$

$$Gain(S_{Rain}, Wind) = Entropy(S) - \sum_{v \in \{Strong, Weak\}} \frac{|S_v|}{|S|} Entropy(S_v)$$

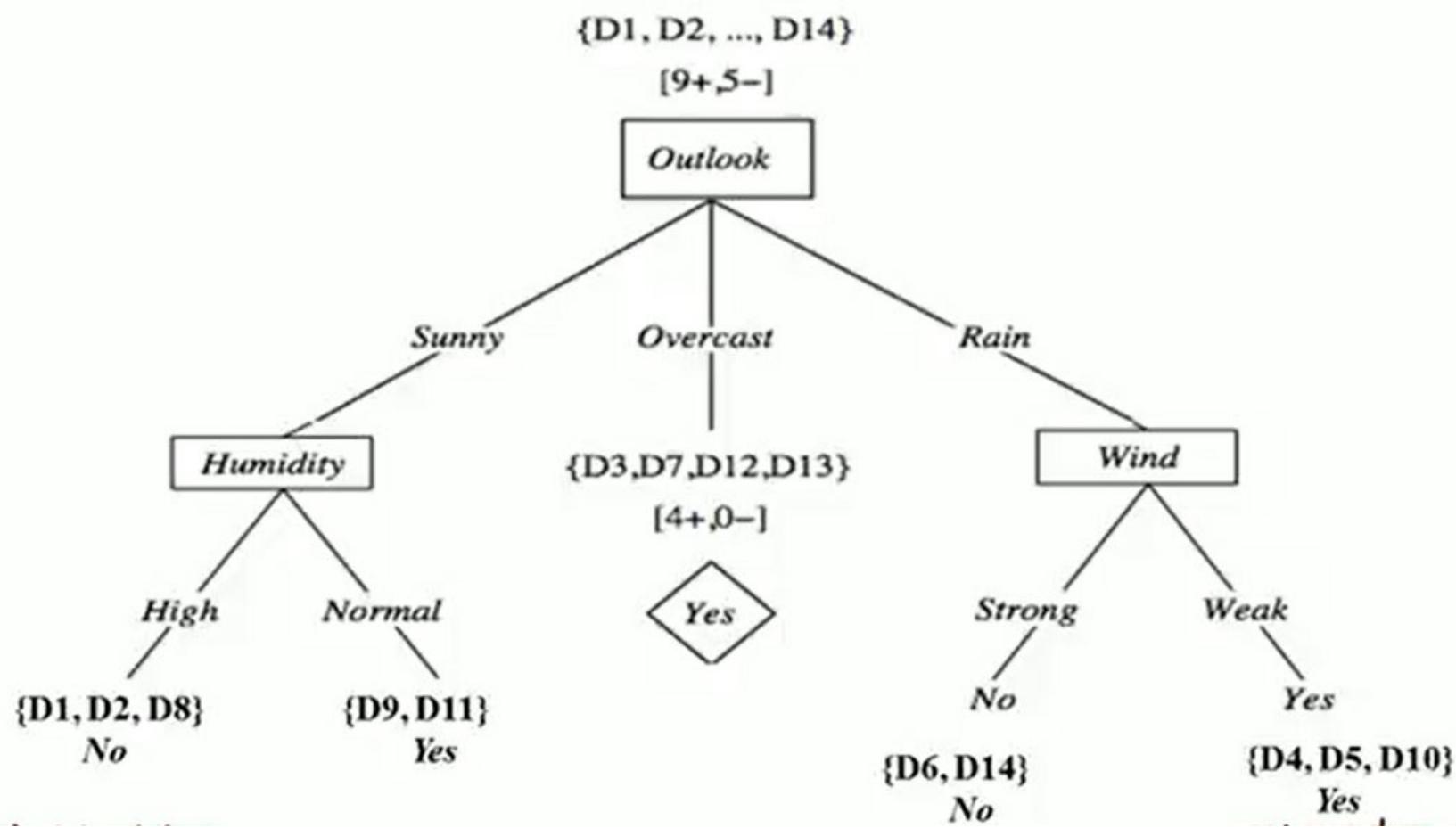
$$Gain(S_{Rain}, Wind) = Entropy(S) - \frac{2}{5} Entropy(S_{Strong}) - \frac{3}{5} Entropy(S_{Weak})$$

$$Gain(S_{Rain}, Wind) = 0.97 - \frac{2}{5} 0.0 - \frac{3}{5} 0.0 = 0.97$$

$$\textit{Gain}(S_{Rain}, \textit{Temp}) = 0.0192$$

$$\textit{Gain}(S_{Rain}, \textit{Humidity}) = 0.0192$$

$$\textit{Gain}(S_{Rain}, \textit{Wind}) = 0.97$$



The End